

Week2 Assignment 2

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Q1. A bank wants to predict whether a loan application will be approved or rejected.

(a) Is this a classification or regression problem?

This is a **classification problem** because the output belongs to one of two categories: **Approved or Rejected**. Classification algorithms are used when the target variable consists of discrete classes.

(b) Which algorithm among Decision Tree, KNN, and Linear Regression would be most suitable?

Among these algorithms, **Decision Tree** is the most suitable because it handles categorical decisions effectively and provides easy-to-understand decision rules. KNN can also be used for classification, but Linear Regression is mainly used for predicting continuous values and is therefore not appropriate for this problem.

Q2. Explain the difference between a feature and a label using a real-world example.

In Machine Learning, **features** are the input variables used to make predictions, while the **label** is the output or target variable that the model tries to predict.

Example: House Price Prediction

Features:

- Area of house
- Number of bedrooms
- Age of house

Label:

- House price

Thus, features represent the independent variables and the label represents the dependent variable.

Q3. A Decision Tree predicts that a customer will purchase a product. Explain how the model reaches this prediction.

A Decision Tree makes predictions by splitting the data into smaller groups based on conditions. Starting from the root node, it asks a series of questions about the customer's characteristics.

For example:

- Is age greater than 30?
- Has the customer purchased previously?
- Is monthly income greater than ₹50,000?

Depending on the answers, the tree follows different branches until it reaches a leaf node. The leaf node contains the final prediction, such as "Customer will purchase the product."

Therefore, the model reaches its prediction by following a sequence of decision rules.

Q4. Why is KNN considered a supervised learning algorithm? Can KNN perform classification if labels are not available?

KNN is considered a **supervised learning algorithm** because it learns from labelled training data. Each training example already has a known class label.

KNN cannot perform classification when labels are not available because it determines the class of a new data point by examining the labels of its nearest neighbors. Without labels, the algorithm has no information to make predictions.

Q5. Suppose a KNN model uses $K = 5$ and the nearest neighbors are:

Yes, Yes, No, Yes, No

What will be the prediction and why?

Number of "Yes" = 3

Number of "No" = 2

Since the majority of the nearest neighbors belong to the class "Yes", the KNN model predicts:

Prediction = Yes

The prediction is based on majority voting among the nearest neighbors.

Q6. Explain the equation of Linear Regression:

$$y = mx + b$$

Where:

- x = Independent variable or feature.
- y = Dependent variable or predicted output.
- m = Slope of the line, which indicates the rate of change of y with respect to x .
- b = Intercept, which is the value of y when $x = 0$.

Example

If

$$y = 5x + 10 \text{ and } x = 2,$$

$$\text{Then } y = 5(2) + 10$$

$$y = 20$$

Thus, Linear Regression represents the relationship between input and output variables using a straight line.

Q7. What is correlation? What do the following values indicate?

Correlation measures the strength and direction of the relationship between two variables.

+1 - A correlation value of +1 indicates a **perfect positive correlation**. As one variable increase, the other variable also increases.

Example: Study hours and marks obtained.

0 - A correlation value of 0 indicates **no relationship** between the variables.

Example: Shoe size and examination marks

-1 - A correlation value of -1 indicates a **perfect negative correlation**. As one variable increase, the other decreases.

Example: Speed and time required to travel a fixed distance.

Q8. A disease detection model produced the following result:

Actual Condition

Disease Present

Prediction

Healthy

(a) What type of prediction is this?

This is called a **False Negative (FN)** prediction.

The patient actually has the disease, but the model predicts that the patient is healthy.

(b) Why can this be dangerous?

False Negative predictions are dangerous because the patient may not receive treatment in time. This can cause the disease to worsen and may even lead to life-threatening situations.

Therefore, minimizing False Negatives is extremely important in medical diagnosis systems.

Q9. What is the purpose of a Confusion Matrix? How is it different from a Correlation Matrix?

Purpose of a Confusion Matrix

A Confusion Matrix is used to evaluate the performance of classification models. It shows:

- True Positives (TP)
- True Negatives (TN)
- False Positives (FP)
- False Negatives (FN)

Using these values, metrics such as accuracy, precision, recall, and F1-score can be calculated.

Difference between Confusion Matrix and Correlation Matrix

Confusion Matrix

Used to evaluate classification models.

Contains TP, TN, FP, FN values.

Helps determine prediction accuracy.

Mainly used in Machine Learning classification.

Correlation Matrix

Used to measure relationships between variables.

Contains correlation coefficients ranging from -1 to +1.

Helps determine how strongly variables are related.

Mainly used in statistical analysis and exploratory data analysis.

Q10. Compare the following algorithms.

Decision Tree - Decision Tree is a supervised learning algorithm that makes predictions by dividing data into branches based on decision rules. It is easy to understand and interpret.

KNN (K-Nearest Neighbors) - KNN predicts the class of a new data point by looking at the classes of its nearest neighbors and using majority voting.

SVM (Support Vector Machine) - SVM separates data into different classes by finding the optimal boundary (hyperplane) that maximizes the distance between classes.

Linear Regression - Linear Regression predicts continuous numerical values by establishing a linear relationship between input variables and output variables.

Neural Network - Neural Networks mimic the functioning of the human brain and consist of interconnected layers of neurons. They are widely used for complex tasks such as image recognition, speech recognition, and deep learning.